

RM uses registers

I = current state

W = code of tape symbols left of tape head (reading right-to-left)

U = code of tape symbols at and right of tape head (reading left-to-right)

Starting with U containing the code of the input string (and all other registers zeroed), the RM program halts if and only if M halts; and in that case I , W and U hold the code of the final configuration.